

CHENG QIAN

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EDUCATION

University of Illinois Urbana Champaign, *Ph.D.*

2024 – Present

- Member of **BLENDER Lab**, advised by Prof. Heng Ji.
- **Research Interest:** Natural Language Processing, LLM Tool Learning and LLM-driven AI Agent; LLM Efficient Training and Knowledge Acquisition; LLM Reasoning; LLM-User Interaction.

Tsinghua University, *B.Eng.* in Computer Science

2020 – 2024

- Member of **THUNLP**, advised by Prof. Zhiyuan Liu.



PUBLICATIONS & TALKS

2025

- [30] **Qian, C.**, Han, P., Luo, Q., He, B., Chen, X., Zhang, Y., Du, H., Yao, J., Yang, X., Zhang, D., et al. EscapeBench: Pushing Language Models to Think Outside the Box. *ACL 2025 Main Conference*.
- [29] **Qian, C.***, Acikgöz, E.C.*, Wang, H., Chen, X., Sil, A., Hakkani-Tür, D., Tür, G., Ji, H. SMART: Self-Aware Agent for Tool Overuse Mitigation. *Findings of ACL 2025*.
- [28] **Qian, C.**, Du, H., Wang, H., Chen, X., Zhang, Y., Sil, A., Zhai, C., McKeown, K. and Ji, H. ModelingAgent: Bridging LLMs and Mathematical Modeling for Real-World Challenges. *Findings of EMNLP 2025*.
- [27] **Qian, C.**, Acikgoz, E. C., He, Q., Wang, H., Chen, X., Hakkani-Tür, D., Tur, G., Ji, H. ToolRL: Reward is All Tool Learning Needs. *arXiv:2504.13958*.
- [26] **Qian, C.**, Liu, Z., Prabhakar, A., Liu, Z., Zhang, J., Chen, H., Ji, H., Yao, W., Heinecke, S., Savarese, S., Xiong, C., Wang, H., et al. UserBench: An Interactive Gym Environment for User-Centric Agents. *arXiv:2507.22034*.
- [25] He, B.*, Ding, N.*, **Qian, C.***, Deng, J., Cui, G., Yuan, L., Gao, H., Chen, H., Liu, Z., et al. Zero-shot Generalization During Instruction Tuning: Insights from Similarity and Granularity. *Findings of ACL 2025*.
- [24] Chen, X.*, Wang, S.*, **Qian, C.***, Wang, H.*, Han, P., Ji, H. DecisionFlow: Advancing Large Language Model as Principled Decision Maker. *Findings of EMNLP 2025*.
- [23] Han, P.*, **Qian, C.***, Chen, X., Zhang, Y., Zhang, D., Ji, H. Internal Activation as the Polar Star for Steering Unsafe LLM Behavior. *Findings of EMNLP 2025*.
- [22]. Wang, H.*, **Qian, C.***, Li, M., Qiu, J., Xue, B., Wang, M., Ji, H., Wong, K. Toward a Theory of Agents as Tool-Use Decision-Makers. *arXiv:2506.00886*.

- [21] Acikgoz, E. C.*, **Qian, C.***, Wang, H.*, Dongre, V., Chen, X., Ji, H., Hakkani-Tür, D., Tur, G. A Desideratum for Conversational Agents: Capabilities, Challenges, and Future Directions. *arXiv:2504.16939*.
- [20] Lu, Y., Yang, S., **Qian, C.**, Chen, G., Luo, Q., Wu, Y., Wang, H., Cong, X., Zhang, Z., et al. Proactive Agent: Shifting LLM Agents from Reactive Responses to Active Assistance. *ICLR 2025*.
- [19] Yang, R.*, Chen, H.*, Zhang, J.*, Zhao, M.*, **Qian, C.**, Wang, K., Wang, Q., et al. EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents. *ICML 2025*.
- [18] Zhu, K., Du, H., Hong, Z., Yang, X., Guo, S., Wang, Z., Wang, Z., **Qian, C.**, Tang, X., et al. MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents. *ACL 2025 Main Conference*.
- [17] Tian, R., Li, Y., Fu, Y., Deng, S., Luo, Q., **Qian, C.**, Wang, S., Cong, X., Zhang, Z., et al. Distance between Relevant Information Pieces Causes Bias in Long-Context LLMs. *Findings of ACL 2025*.
- [16] Zhang, Y., Li, S., **Qian, C.**, Liu, J., Yu, P., Han, C., Fung, Y.R., McKeown, K., Zhai, C., et al. The Law of Knowledge Overshadowing: Towards Understanding, Predicting, and Preventing LLM Hallucination. *Findings of ACL 2025*.
- [15] Reddy, R. G., Dixit, T., Qin, J., Qian, C., Lee, D., Han, J., Small, K., Fan, X., Sarikaya, R., Ji, H. WINELL: Wikipedia Never-Ending Updating with LLM Agents. *Findings of EMNLP 2025*.
- [14] Wu, S., Fung, M., **Qian, C.**, Kim, J., Hakkani-Tür, D., Ji, H. Aligning LLMs with Individual Preferences via Interaction. *COLING 2025*.
- [13] He, B.*, Zhang, W.*, Song, J., **Qian, C.**, Fu, Z., Sun, B., Ding, N., Hong, H., Huang, L., Xue, H., et al. AIR: A Systematic Analysis of Annotations, Instructions, and Response Pairs in Preference Dataset. *COLM 2025*.
- [12] Chen, X.*, Li, G.*, Wang, Z.*, Jin, B., **Qian, C.**, Wang, Y., Wang, H., Zhang, Y., Zhang, D., Zhang, T., et al. RM-R1: Reward Modeling as Reasoning. *arXiv:2505.02387*.
- [11] Wang, H., **Qian, C.**, Zhong, W., Chen, X., Qiu, J., Huang, S., Jin, B., Wang, M., Wong, K., Ji, H. OTC: Optimal Tool Calls via Reinforcement Learning. *arXiv:2504.14870*.
- [10] Wu, S., **Qian, C.**, Fung, Y. R., Liang, P. P., Ji, H. Alice: Proactive Learning with Teacher’s Demonstrations for Weak-to-Strong Generalization. *arXiv:2504.07316*.
- [9] Gao, H., Geng, J., Hua, W., Hu, M., Juan, X., Liu, H., Liu, S., Qiu, J., Qi, X., et al. A Survey of Self-Evolving Agents: On Path to Artificial Super Intelligence. *arXiv:2507.21046*.

[8] **Qian, C.***, He, B.*, Zhuang, Z., Deng, J., Qin, Y., Cong, X., Lin, Y., Zhang, Z., Liu, Z., et al. Tell Me More! Towards Implicit User Intention Understanding of Language Model Driven Agents. *ACL 2024 Main Conference*.

[7] **Qian, C.**, Zhao, X., Wu, S.T. "Merge Conflicts!" Exploring the Impacts of External Distractors to Parametric Knowledge Graphs. *COLM 2024*.

[6] **Qian, C.**, Xiong, C., Liu, Z., Liu, Z. Toolink: Linking Toolkit Creation and Using Through Chain-of-Solving on Open-Source Model. *NAACL 2024 Main Conference*.

[5] **Qian, C.***, Liang, S.*, Qin, Y., Ye, Y., Cong, X., Lin, Y., Wu, Y., Liu, Z., Sun, M. Investigate-Consolidate-Exploit: A General Strategy for Inter-Task Agent Self-Evolution. *arXiv:2401.13996*.

2023

[4] **Qian, C.**, Han, C., Fung, Y., Qin, Y., Liu, Z., Ji, H. Creator: Tool Creation for Disentangling Abstract and Concrete Reasoning of Large Language Models. *Findings of EMNLP 2023*.

[3] Qin, Y.*, **Qian, C.***, Han, X., Lin, Y., Wang, H., Xie, R., Liu, Z., Sun, M., Zhou, J. Recyclable Tuning for Continual Pre-training. *Findings of ACL 2023*.

[2] Qin, Y., Hu, S., Lin, Y., Chen, W., Ding, N., Cui, G., Zeng, Z., Huang, Y., Xiao, C., et al. Tool Learning with Foundation Models. *ACM Computing Surveys*.

2022

[1] Qin, Y.*, **Qian, C.***, Yi, J.*, Chen, W., Lin, Y., Han, X., Liu, Z., Sun, M., Zhou, J. Exploring Mode Connectivity for Pre-trained Language Models. *EMNLP 2022 Main Conference*.

(* indicates equal contribution)

Patents:

1. Name: Training Method, Device, Medium, and Program Product for Predictive Model. Type: Invention. Inventors: Zhiyuan Liu, Maosong Sun, **Cheng Qian (Technical Lead)**, Yujia Qin, Yankai Lin, Ruobing Xie, Jie Zhou, Tsinghua University & Tencent Inc. Ref. Num: CN202211242596.7.

Talks:

1. Towards Reasoning as Action. Invited guest lecture at Northwestern University (CS 496 Agent AI). Host: Manling Li, 2025.5.
2. Towards Reasoning as Action. Invited talk at IBM Research. Host: Avirup Sil, 2025.5.
3. ToolRL: Reward is all Tool Learning Needs. Invited talk at NICE (NLP Academic Exchange Platform). Host: Hongru Wang, 2025.5.
4. Implicit User Intention Understanding of LLM Agent. Invited talk at DEFIRST Reading Group. Host: Rebecca Salganik, 2025.5.

5. Tool Learning and Tool Creation of LLM. Invited talk at SenseTime Smart City Group. Host: Hangyu Mao, 2023.6.



RESEARCH & WORK

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| 1. Summer Research Intern , Salesforce AI Research. Manager: Huan Wang | 2025 |
| Research Direction: Reinforcement learning of LLM agent. | |
| 2. Visiting Student , WInE Lab, CMU. Advisor: Prof. Sherry Wu | 2023 |
| Research Direction: LLM knowledge conflict. | |
| 3. Visiting Student , BLENDER Lab, UIUC. Advisor: Prof. Heng Ji | 2023 |
| Research Direction: LLM agent tool creation. | |
| 4. Undergrad Research Intern , CMU. Advisor: Prof. Chenyan Xiong | 2023 |
| Research Direction: Open-source adaptation of LLM tool use ability. | |



AWARDS

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| 1. Capital One ASKS Research Fellowship, UIUC. | 2025 |
| 2. Comprehensive Excellence Scholarship, Tsinghua University (1/180). | 2021, 2023 |
| 3. Volunteering & Social Excellence Scholarship, Tsinghua University (1/180). | 2022 |
| 4. Awards of Excellent Student Cadre, Tsinghua University. | 2021 |
| 5. Honorable Mention in 2023 Mathematical Contest in Modeling (MCM). | 2023 |



SKILLS & SERVICE

English Skills

- TOEFL (Best) 115/120 (Reading 30, Listening 30, Speaking 26, Writing 29).
- GRE Verbal Reasoning 162/170, Quantitative Reasoning 170/170, Analytical Writing 4/6.

Technical Skills

- Proficient in C/C++, Python (PyTorch), LaTeX, Linux.
- Familiar with various neural networks and state-of-the-art deep learning techniques.

Service

- **Area Chair:** ACL 2025, EMNLP 2025
- **Reviewer:** ARR 2023/2024/2025, ACL 2023/2024, EMNLP 2023/2024, NAACL 2023/2024, COLM 2024/2025, COLING 2025, NeurIPS 2025, AAAI 2025